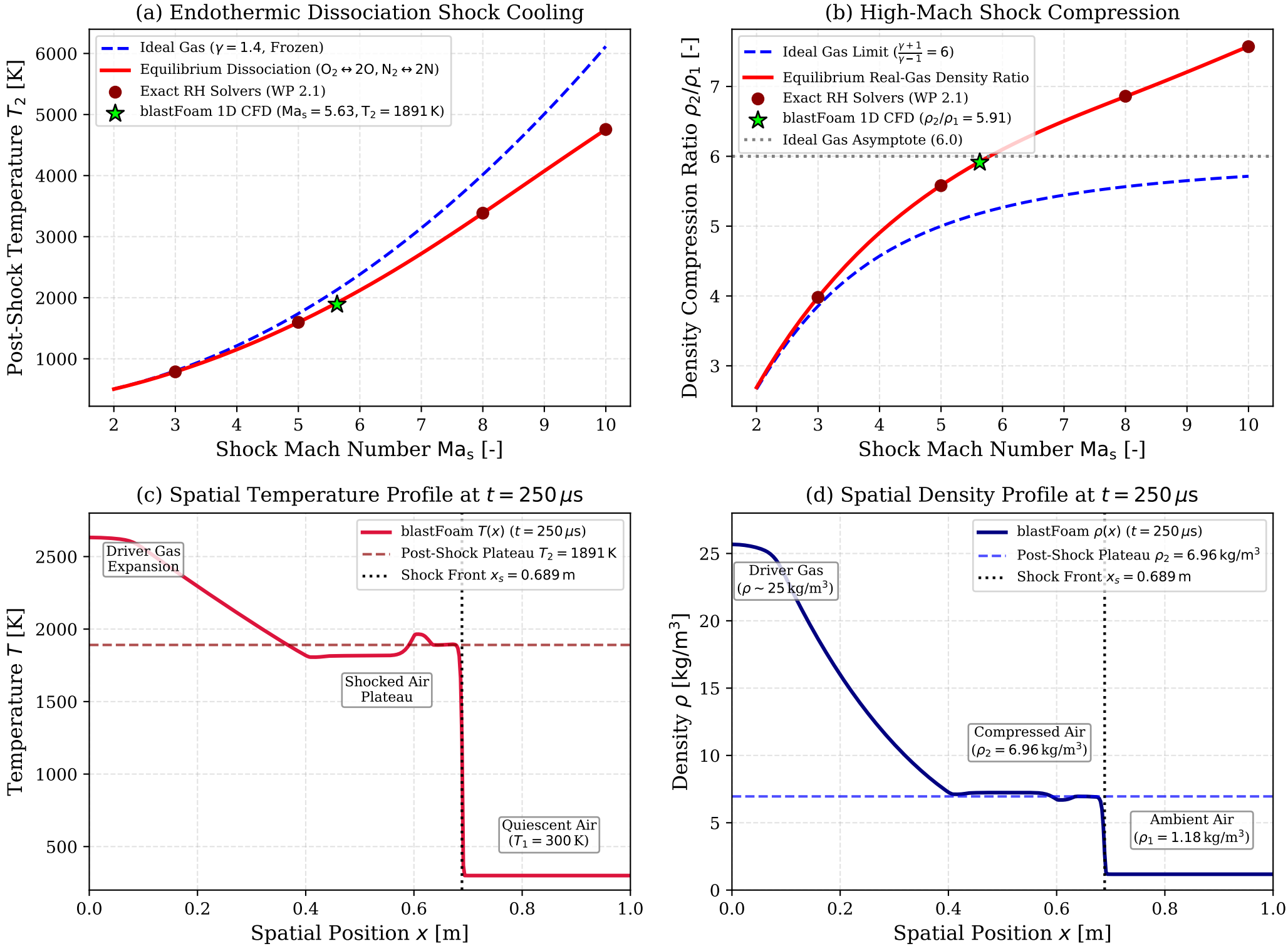
(a) Endothermic Dissociation Shock Cooling
Ideal Gas ($\gamma = 1.4$, Frozen)
Equilibrium Dissociation ($O_2 \leftrightarrow 2O$, $N_2 \leftrightarrow 2N$)
Exact RH Solvers (WP 2.1)
blastFoam 1D CFD ($Ma_s = 5.63$, $T_2 = 1891\,K$)
Post-Shock Temperature T_2 [K]
Shock Mach Number Ma_s [-]
(b) High-Mach Shock Compression
Ideal Gas Limit ($\frac{\gamma+1}{\gamma-1} = 6$)
Equilibrium Real-Gas Density Ratio
Exact RH Solvers (WP 2.1)
blastFoam 1D CFD ($\rho_2/\rho_1 = 5.91$)
Ideal Gas Asymptote (6.0)
Density Compression Ratio ρ_2/ρ_1 [-]
Shock Mach Number Ma_s [-]
(c) Spatial Temperature Profile at $t = 250\,\mu s$
blastFoam $T(x)$ ($t = 250\,\mu s$)
Post-Shock Plateau $T_2 = 1891\,K$
Shock Front $x_s = 0.689\,m$
Driver Gas Expansion
Shocked Air Plateau
Quiescent Air ($T_1 = 300\,K$)
Temperature T [K]
Spatial Position x [m]
(d) Spatial Density Profile at $t = 250\,\mu s$
blastFoam $\rho(x)$ ($t = 250\,\mu s$)
Post-Shock Plateau $\rho_2 = 6.96\,kg/m^3$
Shock Front $x_s = 0.689\,m$
Driver Gas ($\rho \sim 25\,kg/m^3$)
Compressed Air ($\rho_2 = 6.96\,kg/m^3$)
Ambient Air ($\rho_1 = 1.18\,kg/m^3$)
Density ρ [kg/m^3]
Spatial Position x [m]